through P is made of all the points in the plane for which y=Py.

So, when we split the plane along a vertical line through P=(Px,Py), what we really mean is that we create two partitions, one for the points L in the plane for which Lx<Px, and one for those points R for which Rx>Px. And similarly, for horizontal lines, using the y coordinates of the points.

This binary partitioning allows us to use binary trees to index our points: each node in the tree is the pivot we chose to partition the remaining region of space, and its left and right subtrees gather all points in the two partitions, and represent the two sub-regions resulting from the split we perform along the pivot (check out figure 9.2 and figure 9.5).

Can this algorithm be generalized to higher dimensions? Yes, it naturally allows a generalization to higher dimensions, because we split on a single coordinate for each point, but we can do a round-robin through all the coordinates of a k-dimensional space, and the i-th level of the binary tree we build will correspond to a split along the (i mod k)-th dimension.

Meaning that in a 2-D space, the root will split the plane along the x axis, its children will split each of the two semi-planes along the y axis, then their children again along the x axis and so on and so forth. In a k-dimensional space, with k > 2, we start with the first coordinate at level 0 (the root), then move to the second at height 1, the third at height 2 etc...